

MODULE 3A • EPISODE 1

Making Measurements

Foundations of Science Measurements

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Foundations of Science Measurements

This episode is about the most basic act in experimental science: making a measurement and writing it down honestly. The panel starts with the difference between qualitative and quantitative observations, then takes apart a scientific measurement into its three parts — a best guess, a tolerance, and a unit — and explains why a measurement is really a range of values and a probability statement. From there the discussion turns to the measuring devices themselves. Digital devices, like the digital caliper, take a tolerance equal to their least count. Analog devices, like the micrometer and the metric ruler, usually take half their least count. The episode closes with beam balances and graduated cylinders, and with the point that careful technique can sometimes get more precision out of an instrument than its markings suggest.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist specializing in precision measurements

Learning Objectives

After this episode you should be able to:

- ◆ Distinguish qualitative from quantitative observations and give an example of each.
- ◆ Name the three parts of a scientific measurement and write a measurement correctly.
- ◆ Convert a measurement such as 12.34 ± 0.02 cm into its confidence interval.
- ◆ Explain why every scientific measurement is a probability statement.
- ◆ Explain why a tolerance that is too large makes a measurement meaningless.
- ◆ Distinguish digital from analog devices and find the least count of each.
- ◆ Apply the 100% rule to digital devices and the one-half rule to analog devices.
- ◆ Write correctly formatted measurements for a digital caliper, micrometer, ruler, beam balance, and graduated cylinder.

QUICK REFERENCE

Key Terms

Term	Definition
Qualitative observation	A descriptive observation that involves no numbers, such as “the solution turns bright blue.”
Quantitative observation	An observation that involves numbers, such as a temperature of 92 °C.
Scientific measurement	A measurement written in three parts: best guess, tolerance, and unit, as in 12.34 ± 0.02 cm.
Best guess	The central value of a measurement; also called the best estimate.
Tolerance (ILE)	The $\pm$ part of a measurement; also called the instrument limiting error.
Unit	The quantity a measurement is expressed in, such as centimeters or grams.
Confidence interval	The range of values from best guess minus tolerance to best guess plus tolerance, endpoints included.
True value	The actual value of the quantity being measured, which can never be known with certainty.
Least count	The smallest scale division of a measuring device; the smallest step its reading can change by.
Least significant digit	On a digital display, the rightmost digit — the one that changes by the smallest amount.
Significant figures	The digits in a measurement that carry meaning about its precision.
Digital device	A measuring device that shows its reading as a sequence of digits on a screen.
Analog device	A measuring device with a ruled surface that is lined up with an edge, needle, or other marking.
Digital caliper	A digital instrument that measures lengths, inside diameters, and depths to 0.01 mm.
Zero error	An error introduced when a device does not read zero when it should, such as caliper jaws that are closed.
Micrometer caliper	An analog instrument — essentially a precision screw attached to a ruler — that measures lengths to 0.01 mm.
100% rule	Taking the tolerance equal to the least count; used for digital devices.
One-half rule	Taking the tolerance equal to half the least count; usually used for analog devices. Also called the 50% rule.
Meniscus	The curved surface a liquid forms where it meets the wall of its container.
Parallax	An apparent shift in a reading caused by viewing a scale from an angle rather than straight on.

Term	Definition
Standard deviation of the mean	A statistically determined tolerance used when many identical measurements have been made.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
3	Parts of a scientific measurement: best guess, tolerance, unit
12.34 ± 0.02 cm	The episode's example measurement; its confidence interval is 12.32 to 12.36 cm
100%	Probability assigned to the confidence interval of a single measurement
~100	Number of identical measurements at which statistics can improve the best guess and tolerance
64 - 66 in	Confidence interval for a height of 5 ft 5 in ± 1 in
0.01 mm	Least count and tolerance of the digital caliper
0.1 mm	Rough thickness of one sheet of paper (10 sheets ≈ 1 mm)
0 - 50	Numbers on one full turn of the micrometer thimble
0.50 mm	How far the micrometer spindle moves in one full turn of the thimble
0.005 mm	Tolerance of the micrometer (half its 0.01 mm least count)
0.5 mm	Tolerance of a metric ruler marked in millimeters
0.1 g	Typical least count and tolerance of a beam balance
1 mL = 1 cm³	A milliliter is exactly one cubic centimeter
0.2 mL	Tolerance of a graduated cylinder read carefully at the top of the meniscus

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Observations and the Three-Part Measurement

Science uses two kinds of observations. **Qualitative observations** are descriptive and involve no numbers: a solution turns bright blue, a material feels rough. **Quantitative observations** always involve numbers: instead of “hot,” a temperature of 92 °C. Numbers are what make science reproducible and reliable.

Every **scientific measurement** has three parts, two of them numbers. In 12.34 ± 0.02 cm, 12.34 is the **best guess** or best estimate; ± 0.02 is the **tolerance**, or **instrument limiting error (ILE)**; and cm is the **unit**.

“Plus or minus — which is it?” The answer is to do both. Subtracting and adding the tolerance gives a **confidence interval**, here 12.32 to 12.36 cm, with both endpoints included. It is perfectly acceptable to write the measurement as that interval.

Measurements are written as ranges because no measurement can be made with absolute certainty. Instead, a probability is assigned to the interval — for example 67% or 100% — meaning the interval captures the **true value** with that confidence. Every scientific measurement is therefore a **probability statement**. For a single measurement of 12.34 ± 0.02 cm, we are 100% sure the true value lies between 12.32 and 12.36 cm.

Part 2 — Choosing the Right Tolerance

For a **single measurement**, the goal is the **smallest tolerance** that still gives a confidence interval with a probability of 100%. If many identical measurements are made — on the order of 100 — statistics can do better: the arithmetic average becomes the best guess, and the **standard deviation of the mean** becomes the tolerance.

The carnival example shows why the tolerance matters. A height guesser who calls someone 5 ft 5 in ± 1 ft has said almost nothing; the customer wants her dollar back. Calling it 5 ft 5 in ± 1 in, when her true height is 5 ft 6 in, earns the dollar: it says her height lies between 64 and 66 inches, inclusive, with 100% certainty, and rules out 63 or 67 inches.

Carnival guessers have a trick: for most people, height is about equal to **arm span**, the distance from fingertip to fingertip with the arms stretched out, and arm span is easy to judge by eye.

So how is the smallest 100% tolerance found for a single measurement? By examining the measuring device itself.

Part 3 — Digital Devices: The Digital Caliper

A **digital device** shows its reading as a sequence of digits on a screen. The rightmost digit, the one that changes by the smallest amount, is the **least significant digit**. A **digital caliper** reading 20.14 mm has precision to a hundredth of a millimeter and four **significant figures**; its least significant digit is the 4.

The digital caliper is a glorified ruler. A millimeter is about the thickness of 10 sheets of paper, so 0.01 mm is a tenth of one sheet — far finer than anyone can see on an ordinary ruler. Calipers read in millimeters or inches and make external measurements with the external jaws, internal measurements with the internal jaws, and depth measurements with the depth probe. Their analog cousin is the **Vernier caliper**.

With the jaws closed, the display must read 0.00 mm. If it does not, press the tare or zero button; otherwise every measurement carries a **zero error**.

The tolerance of any device is chosen from its **least count**, its smallest scale division, as a fraction of the least count up to 100%. The caliper's reading can only step by 0.01 mm, so its least count is 0.01 mm. For digital devices the tolerance is **100% of the least count**, because repeated readings vary by about 0.01 mm and the last digit often flickers up or down. A caliper measurement is written like 12.34 ± 0.01 mm — with the best guess and the tolerance carrying the same number of decimal places.

Part 4 — Analog Devices: The Micrometer and the One-Half Rule

An **analog device** has a ruled surface that is lined up with an edge, needle, or other marking. The **micrometer caliper** — a “mike” — is essentially a glorified screw attached to a ruler.

Turning the **thimble** counterclockwise, as seen from the ratchet end, moves the **spindle** away from the fixed **anvil**. One full turn runs through the numbers 0 to 50 on the thimble and advances the spindle exactly 0.50 mm, which shows on the **sleeve's** ruler, marked in half millimeters. Two full turns move it 1 mm. The reading is taken on the sleeve's center line, and the least count is 0.01 mm, the space between adjacent thimble markings.

Analog devices usually follow the **one-half rule**: the tolerance is half the least count. For the micrometer that is 0.005 mm, so a measurement reads like 15.500 ± 0.005 mm or 15.505 ± 0.005 mm.

The reasoning is conservative: we can see two parts in the least count. With 100% confidence, the sleeve's center line is either right on a thimble marking or somewhere between two adjacent ones. Adding and subtracting half the least count centers a window one least count wide around the best guess.

Part 5 — Rulers, Balances, and Graduated Cylinders

A **metric ruler** marked in millimeters is analog, with a least count of 1 mm. By the one-half rule its tolerance is 0.5 mm, so measurements look like 13.0 ± 0.5 mm, 13.5 ± 0.5 mm, or 14.0 ± 0.5 mm. The last digit of both numbers can only be 0 or 5.

A **beam balance** is also analog, but it is less precise than a micrometer and takes the **100% rule**. Its least count is typically 0.1 g, so its tolerance is 0.1 g, as in 45.3 ± 0.1 g.

A **graduated cylinder** measures liquid volume in milliliters; a milliliter is exactly one cubic centimeter, about the volume of one gram of water at 4 °C. With a least count of 1 mL, the one-half (50%) rule would give ± 0.5 mL. But water creeps up the glass and forms a curved **meniscus**. Since most measurements involve a change in level, either the top or the bottom of the meniscus can be used, as long as the choice is consistent.

Reading the top of the meniscus straight on, at eye level to avoid **parallax**, one can see five parts in the least count, giving a measurement like 66.4 ± 0.2 mL. The precision you get depends not just on the instrument but on how carefully and skillfully you use it.

The next episode takes up significant figures, precision versus accuracy, and systematic errors.

STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Part 1 — Observations and the Three-Part Measurement

1. What is the difference between a qualitative and a quantitative observation? Give an example of each.

ANSWER A qualitative observation is descriptive and involves no numbers, such as a solution turning bright blue or a material feeling rough. A quantitative observation involves numbers, such as a temperature of 92 °C.

2. Name the three parts of the scientific measurement 12.34 ± 0.02 cm.

ANSWER Best guess (best estimate): 12.34. Tolerance, or instrument limiting error (ILE): ± 0.02 . Unit: cm.

3. What is the confidence interval for 12.34 ± 0.02 cm, and are its endpoints included?

ANSWER 12.32 cm to 12.36 cm. Yes, both endpoints are included.

4. Why are scientific measurements written as ranges rather than single numbers?

ANSWER No measurement can be made with absolute certainty. Writing a range lets us say, with an assigned probability, that the true value lies somewhere in it.

5. In what sense is every scientific measurement a probability statement?

ANSWER We can never know the true value for certain, but we can state the probability — for a single measurement, 100% — that it lies between the endpoints of the confidence interval.

Part 2 — Choosing the Right Tolerance

1. For a single measurement, what tolerance should we look for?

ANSWER The smallest tolerance that, added to and subtracted from the best guess, gives a confidence interval with a probability of 100%.

2. If about 100 identical measurements are made, what replaces the best guess and the tolerance?

ANSWER The arithmetic average becomes the best guess, and the standard deviation of the mean becomes the tolerance.

3. Why would a customer want her dollar back if her height were guessed as 5 ft 5 in ± 1 ft?

ANSWER The tolerance is so large that the measurement is crude and imprecise to the point of lacking all meaning — it covers a range of two feet.

4. Write the confidence interval for 5 ft 5 in ± 1 in, in inches. What heights does it rule out?

ANSWER 64 to 66 inches, inclusive. It rules out heights such as 63 or 67 inches.

5. What trick do carnival height guessers use?

ANSWER For most people, height is about equal to arm span, the fingertip-to-fingertip distance with arms outstretched, which is easy to estimate by eye.

Part 3 — Digital Devices: The Digital Caliper

1. What makes a measuring device digital, and what is its least significant digit?

ANSWER It shows its reading as a sequence of digits on a screen. The least significant digit is the rightmost one, which changes by the smallest amount.

2. A digital caliper reads 20.14 mm. How many significant figures does the reading have, and what is its precision?

ANSWER Four significant figures, with precision to one hundredth of a millimeter (0.01 mm).

3. Name the three kinds of measurement a digital caliper can make, and the part used for each.

ANSWER External measurements with the external jaws, internal measurements with the internal jaws, and depth measurements with the depth probe.

4. What should the caliper read with its jaws closed, and what happens if you don't correct it?

ANSWER 0.00 mm. If not, press the tare or zero button; otherwise every measurement includes a zero error.

5. What is the least count of a digital caliper, and what tolerance does it take? Why?

ANSWER Least count 0.01 mm; tolerance 0.01 mm, 100% of the least count. Repeated readings vary by about 0.01 mm, and the last digit often flickers up or down.

6. Why should the best guess and the tolerance have the same number of decimal places?

ANSWER Different numbers of decimal places would imply different levels of precision, which makes no sense for a single measurement.

Part 4 — Analog Devices: The Micrometer and the One-Half Rule

1. What makes a measuring device analog?

ANSWER It has a ruled surface that is lined up with an edge, needle, or other marking.

2. How far does the micrometer spindle move in one full turn of the thimble, and in two?

ANSWER 0.50 mm in one full turn (the thimble runs 0 to 50); 1 mm in two.

3. What is the least count of a micrometer, and what is its tolerance?

ANSWER Least count 0.01 mm, the space between adjacent thimble markings. Tolerance 0.005 mm, by the one-half rule.

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4. State the one-half rule and explain why it is used for analog devices.

ANSWER The tolerance is half the least count. Conservatively, we can see two parts in the least count: with 100% confidence, the reading line is either on a marking or between two adjacent ones, so adding and subtracting half the least count centers a window one least count wide around the best guess.

5. Write a correctly formatted micrometer measurement.

ANSWER For example, 15.500 ± 0.005 mm or 15.505 ± 0.005 mm.

Part 5 — Rulers, Balances, and Graduated Cylinders

1. What is the tolerance of a metric ruler marked in millimeters? What can the last digit of a measurement be?

ANSWER 0.5 mm, half the 1 mm least count. The last digit of both the best guess and the tolerance can only be 0 or 5, as in 13.5 ± 0.5 mm.

2. Which rule does the episode use for a beam balance, and how is a measurement written?

ANSWER The 100% rule: least count 0.1 g, tolerance 0.1 g, as in 45.3 ± 0.1 g.

3. How is a milliliter related to a cubic centimeter?

ANSWER A milliliter is exactly one cubic centimeter — about the volume of one gram of water at 4 °C.

4. What is a meniscus, and which part of it should you read?

ANSWER The curved surface water forms where it creeps up the glass wall. Either the top or the bottom can be used, since most measurements involve a change in level, as long as you use the same one each time.

5. Why should you read a graduated cylinder straight on, at the level of the liquid?

ANSWER To avoid parallax, an apparent shift in the reading caused by viewing the scale from an angle.

6. How can a graduated cylinder give a tolerance of 0.2 mL when its least count is 1 mL?

ANSWER Reading the top of the meniscus carefully and straight on, one can see five parts in the least count, so the tolerance is 1/5 of a milliliter. Careful technique can get more precision than the basic rule suggests.

TEACHER NOTES

Corrections and Cautions

Points in the episode that need a correction, a qualification, or a prepared answer for the question a sharp student will ask.

C1 “67%” should be 68%.

The standard confidence for one standard deviation is about 68% (68.3%). The episode's “67%” is a slip. Students who meet 68% in a later course may think one figure is wrong.

C2 “100% certain” is an idealization.

The instrument limiting error covers reading limits only. A zero error, a miscalibrated instrument, or a parallax mistake can put the true value outside the interval. The next episode's systematic errors are exactly this. Worth saying explicitly so students don't take 100% literally.

C3 Heights are not just whole inches.

The episode says the height is 64, 65, or 66 inches. The interval 64–66 in includes every value in between, such as 64.7 in. Measured quantities like length are continuous; only the readings are discrete.

C4 The beam balance's justification is weak.

The episode calls the beam balance analog but applies the 100% rule “because these scales are less precise than micrometers.” Being less precise is not a reason to change rules. A better reason is that the pointer's swing and friction make readings finer than the least count unreliable. Also note that many classroom balances read to 0.01 g.

C5 Water at 4 °C.

A milliliter is exactly one cubic centimeter by definition. One gram of water at 4 °C occupies 1.00003 cm³, so “the volume of one gram of water” is very close but not exact.

C6 Top versus bottom of the meniscus.

The standard laboratory practice for water is to read the bottom of the meniscus. The episode's preference for the top works because it measures changes in level and is consistent, but students in other lab courses will be told the bottom.

C7 “Tear” button.

The button is the tare (or zero) button. Captions and student notes sometimes spell it “tear.”

C8 Two names for each rule.

The episode uses “one-half rule” and “50% rule” interchangeably, and “100% of the least count” and “100% rule” interchangeably. Students may think there are four rules.

C9 “See five parts in the least count.”

This is consistent with the one-half rule's “two parts”: tolerance equals the size of one visible part. Two parts of 1 mL give 0.5 mL; five parts give 0.2 mL. Making that link explicit helps students generalize.

TEACHER NOTES

Teaching the Episode

Start with “Which is it?”

Write 12.34 ± 0.02 cm on the board and ask the class the episode's question. Having students work out the interval 12.32–12.36 cm themselves makes the confidence-interval idea stick.

Run the carnival experiment.

Pair students to measure height and arm span with a meter stick and tape. Plotting arm span against height is a quick, memorable data set, and each measurement must be written with its tolerance.

Pass around real instruments.

If you have a digital caliper and a micrometer, let students zero the caliper, watch the last digit flicker, and count thimble turns on the micrometer. Two turns for 1 mm is easier to believe after seeing it.

Practice the format.

Give a list of readings — a caliper reading, a micrometer reading, a ruler reading, a balance reading — and have students write each as best guess $\pm$ tolerance with units and matching decimal places. Format errors are the most common mistake.

Set up the next episode.

Correction C2 is a natural bridge to systematic errors. Ask what happens to the “100% sure” interval if the caliper was never zeroed.